Forest Biomass Estimation using Data from Field, Airborne and Spaceborne Lidar

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Outline

- Background
  - Forest biomass estimation using field and lidar data
- Exercise 1
  - GEDI waveform simulation using rGEDI in R
- Exercise 2
  - Development of local AGB model with field AGB and simulated ALS Relatiev Height metrics in R
- Exercise 3
  - AGB mapping in Google Earth Engine

Goal

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- Learn to develop local forest biomass model using field and remote sensing data and its application for AGB mapping.

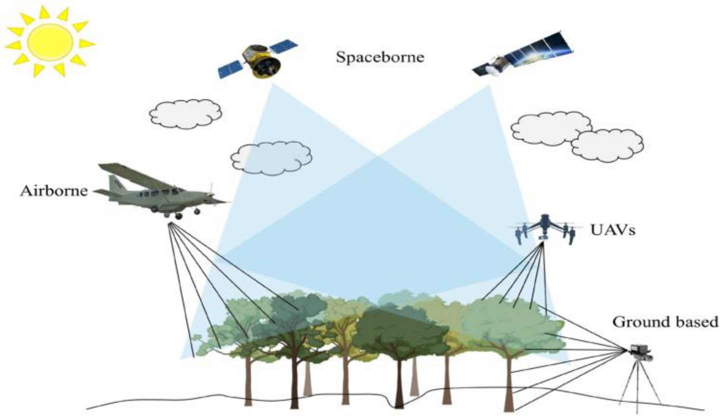
Data for forest biomass estimation/prediction

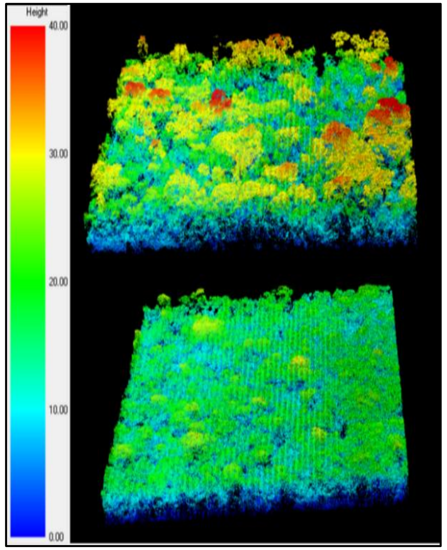
- Forest inventory
- Airborne lidar (ALS)
- Spaceborne lidar (GEDI)

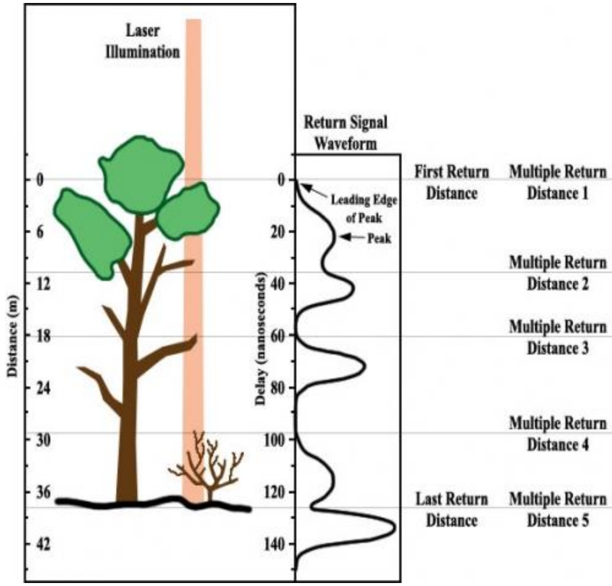
Measuring/estimating forest biomass

- Forest Inventory
  - Take tree measurements such as Diameter at breast height (DBH), canopy height, basal area etc in field.

- LiDAR
  - Indirect measurement of forest structure information such as Canopy Height (CH), Relative Heights (RHs) and others.







Forest biomass estimation in field

- Destructive biomass measurement
  - Harvest sample trees and weight their components (foliage, branches, stem, roots etc).
  - Provide most accurate estimates.
- Forest inventory
  - A process of collecting forest data including tree species, size and other characteristics
  - Diameter at Breast Height (DBH), Tree Height and Basal Area
  - Specie identification
  - Others depending upon interests of users.
- Allometric Equations for forest biomass estimation
  - Mathematical models developed by establishing a relationship between tree biomass and tree data (DBH, height, etc) through regression analysis.

Allometric Models

- Mathematical models to estimate biomass of each tree in field plot based upon physical characteristics of trees such as DBH, height (H_t), wood density (ρ) etc.

- Allometric Equations for tropical forest (Chaves et al., 2004)

$$\textit{Living Tree AGB} = 0.063\ (\rho \times DBH^2 \times H_t)^{0.976}$$

$$\textit{Living Palms AGB} = 0.03781 \times DBH^{2.7483}$$

$$\textit{Living Lianas AGB} = 0.03798 \times DBH^{2.657}$$

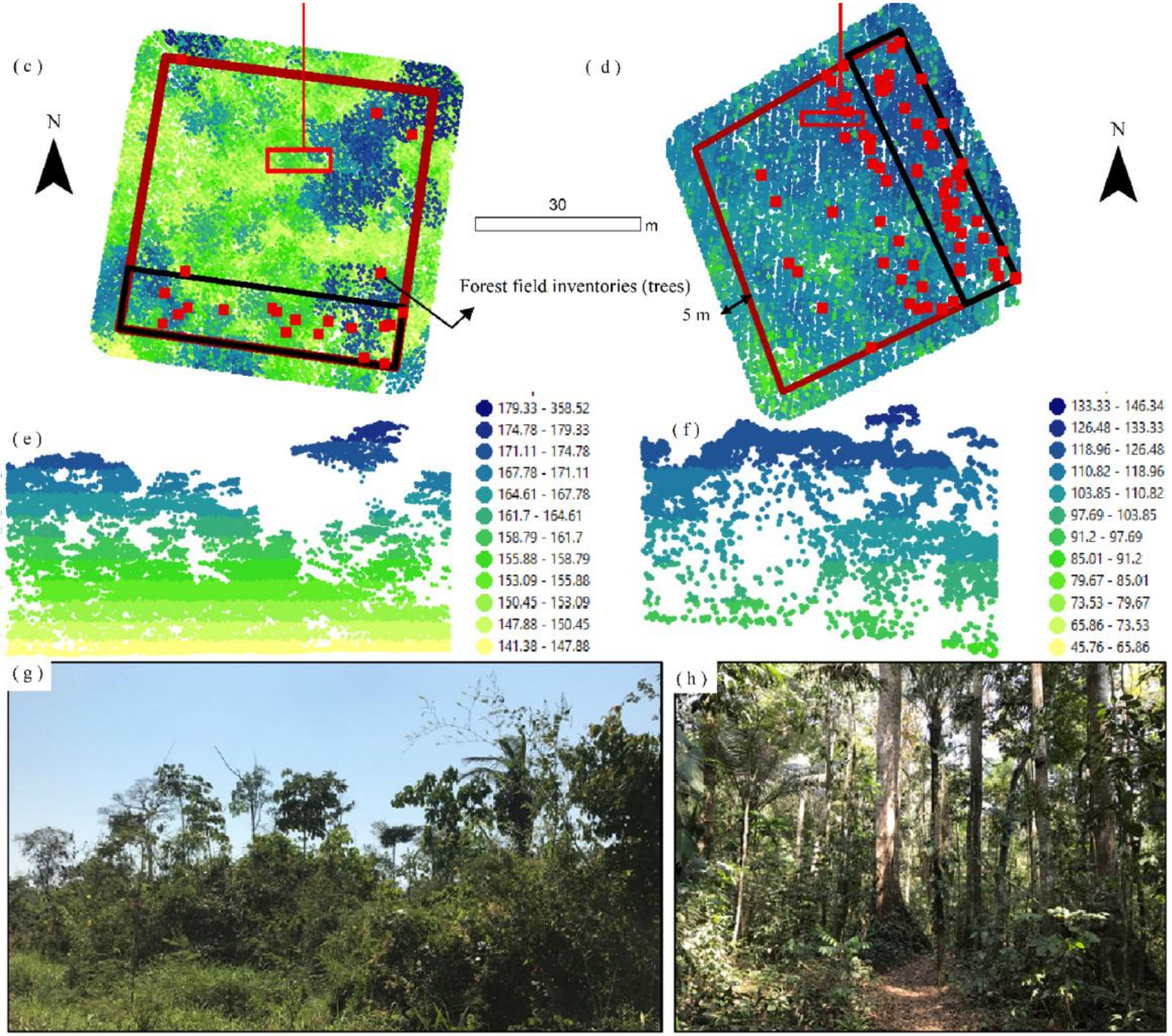
Where ρ is wood density, H_t is tree height.

- Then, calculate total forest biomass in Mg/ha for each field plot.

1 ha-size field plot in Acre

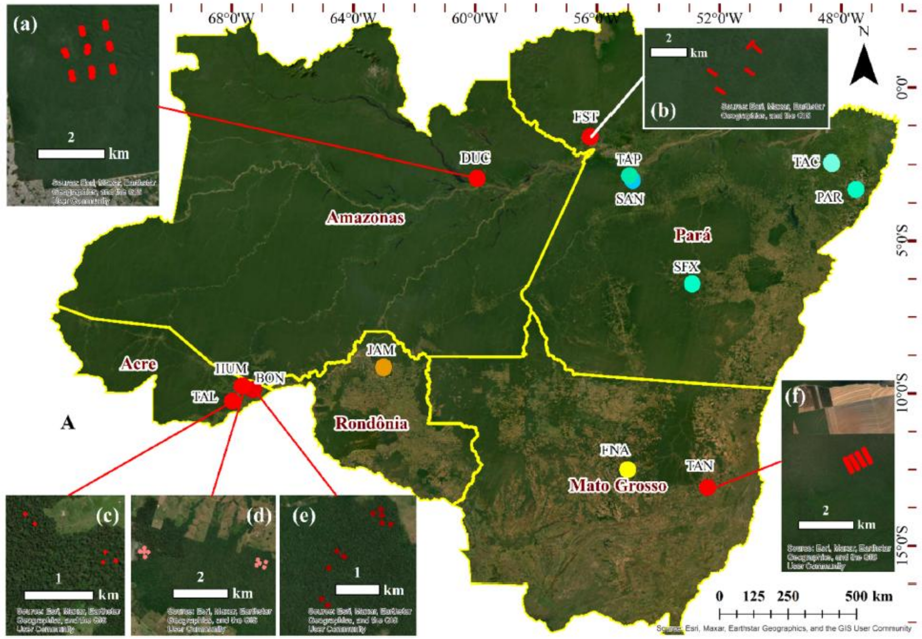
Airborne/UAV lidar

- Discreate Return/Point cloud
- Point density: 4 points/m^2 at minimum
- Peak Return Structure information
- Spatially continuous coverage
- Forest structure information such as canopy height, relative heights, canopy coverage and gaps etc.
- Stratification of vertical forest profile to estimate relative height.

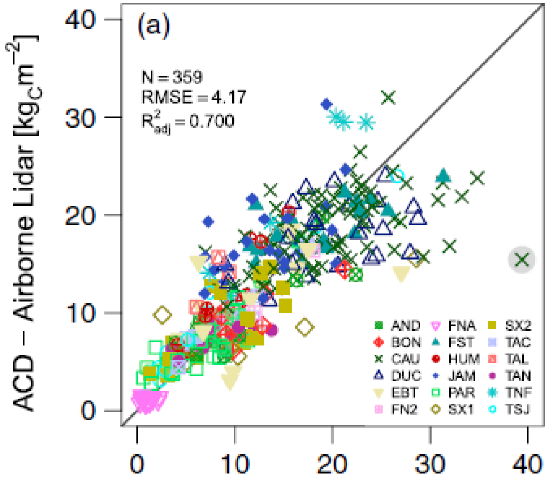


Carbon estimation model using Airborne LiDAR

- Longo et al. (2016), *Global Biogeochemical Cycles*
- AGB models developed from several field plots and ALS data collected in the Amazon by **Sustainable Landscape Project**.
- Parametric and non-parametric (ML) models using parameters derived from ALS data

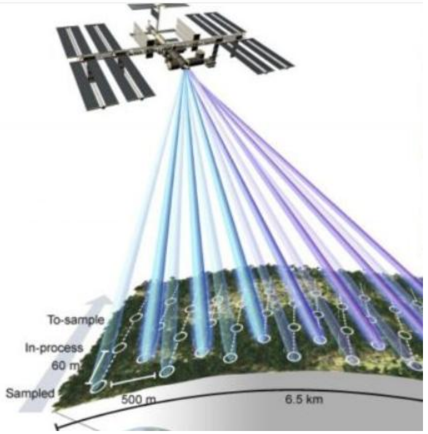


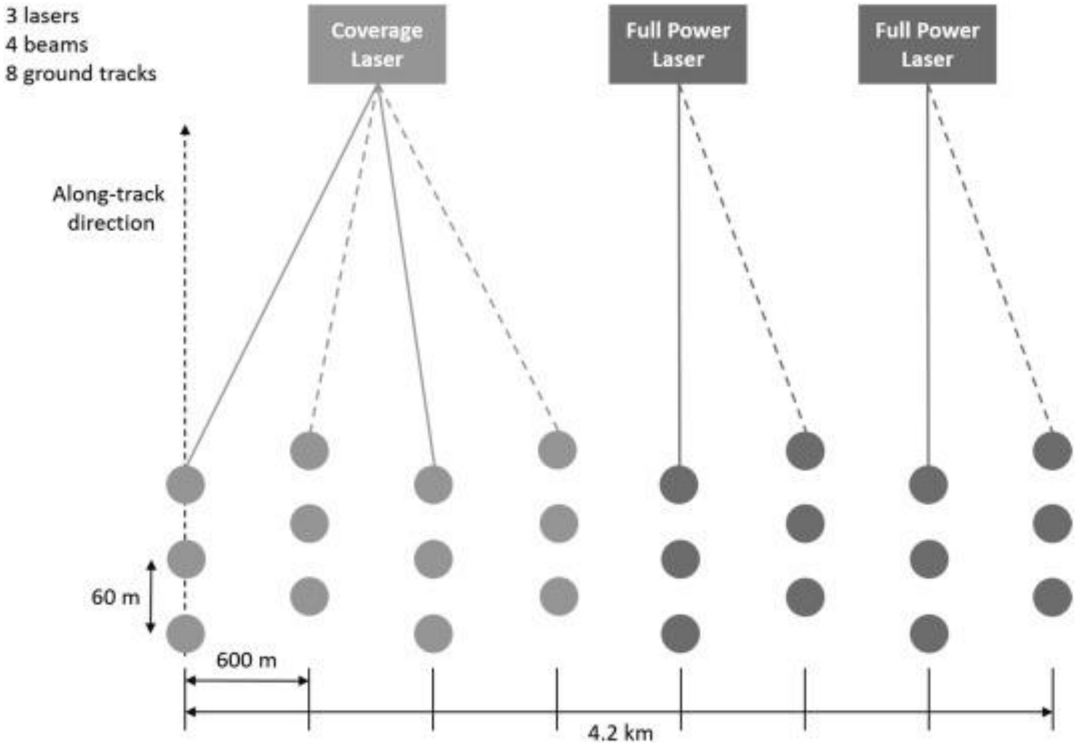
$$\mathrm{ACD}_{\mathrm{ALS}} = 0.20(0.08)\,\bar{h}^{2.02(0.14)}\,\kappa_h^{0.66(0.13)}\,h_5^{0.11(0.04)}\,h_{10}^{-0.32(0.06)}\,h_{\mathrm{IQ}}^{0.50(0.15)}\,h_{100}^{-0.82(0.13)} + E_{\mathcal{N}}\left[\mu = 0, \sigma = 0.66(0.12)\,\mathrm{ACD}_{\mathrm{ALS}}^{0.71(0.08)}\right],$$

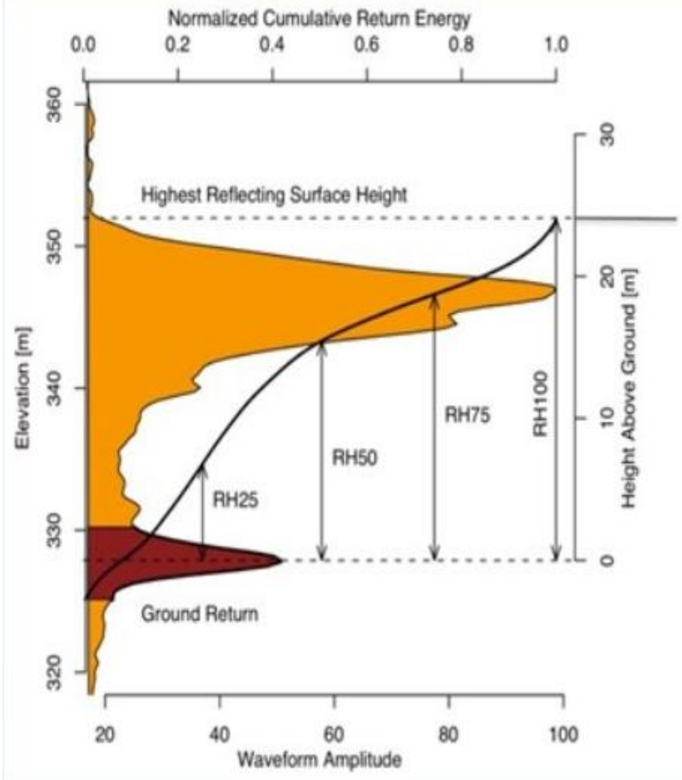


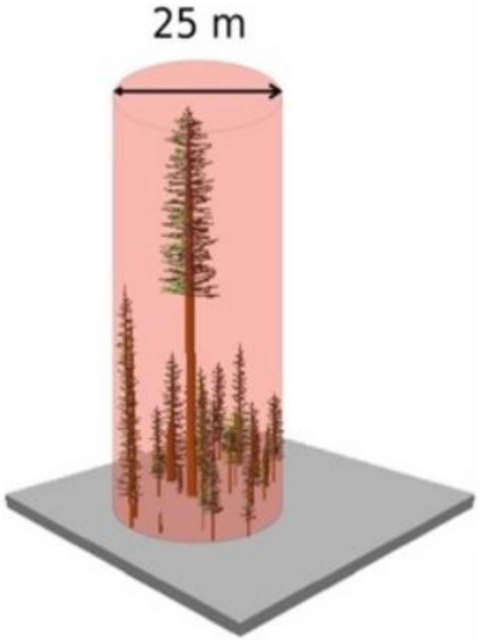
ACD= 0.54 × TCH*1.76

Spaceborne Lidar: Global Ecosystem Dynamics Investigation (GEDI)



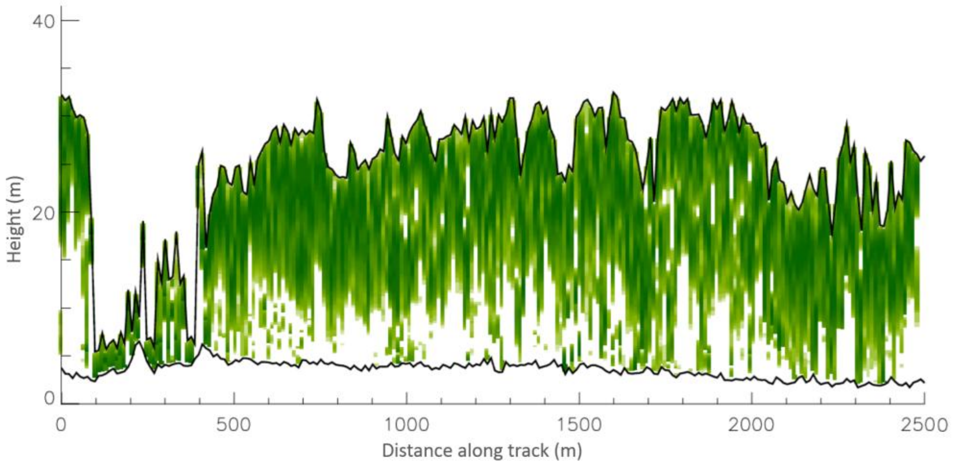




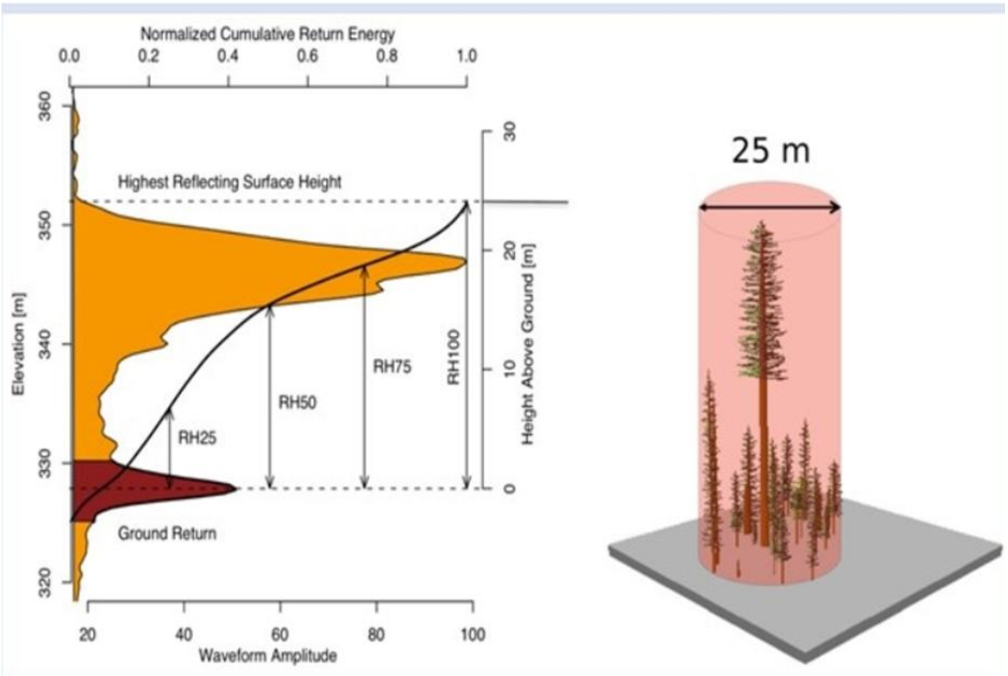


GEDI Products

- L1B: Geolocated Waveforms
- L2A: Ground Elevation and Height Metrics
- L2B: Canopy Cover Fraction, PAI, FHD
- L4A: Footprint level AGBD
- L4B: Gridded AGBD



Along track lidar rerturn energy showing vertical distribution of vegetation
(https://gedi.umd.edu/mission/mission-overview/)

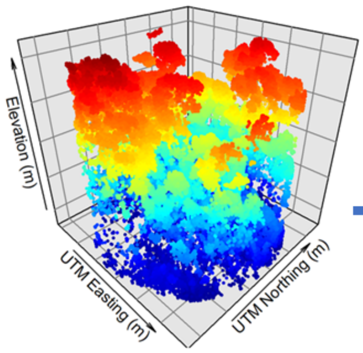


Exercises: Waveform simulation, AGB modeling and mapping

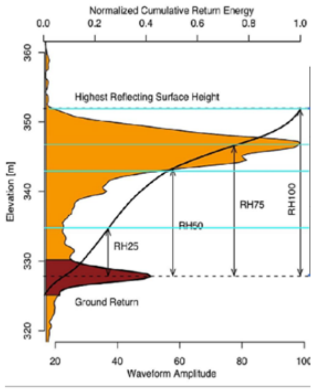
Exercise 1 using R

ALS and GEDI simulation

a) GEDI waveform simulation and metrics extraction from ALS



ALS Point Cloud



Simulated GEDI WV

RH metrics

Exercise 2 using R

Biomass Modeling

b) Field biomass measurements

Field inventory data

Allometric equations

Plot level biomass

c) Biomass modeling

Regression models

Biomass Prediction

Validation

Exercise 3 using GEE

AGBD Map

e) Prediction of biomass at GEDI footprint level using L2A RH data

f) Biomass Mapping by fusing GEDI-footprint level biomass with other sensors

g) Uncertainty Analysis

Local AGB model

Field measurements and ALS (UAV) lidar data collection

GEDI full-waveform simulation from ALS

Extraction of Relative Height Metrics from simulated GEDI WVs

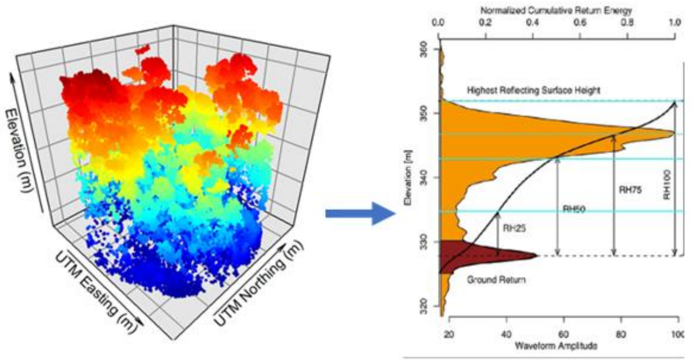
Development of local AGB models using field AGB and RH metrics

Validation

Exercise 1 using R

ALS and GEDI simulation

a) GEDI waveform simulation
and metrics extraction from ALS

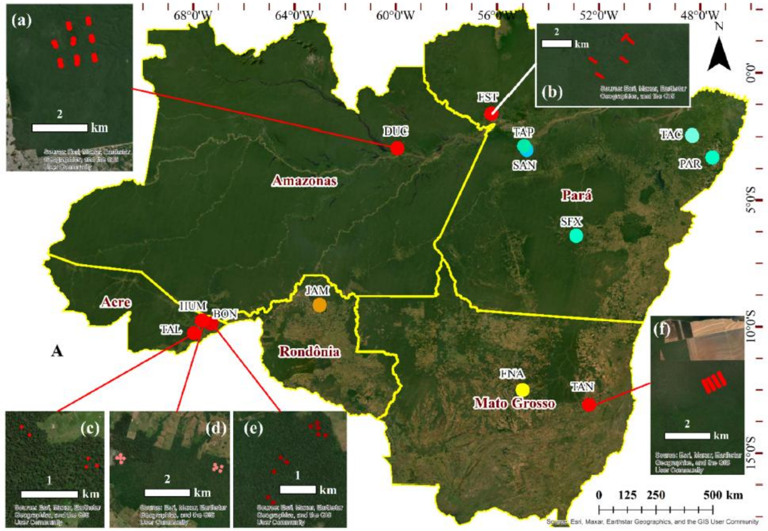


ALS Point Cloud

Simulated GEDI WV

RH metrics

ML model for forest biomass estimation



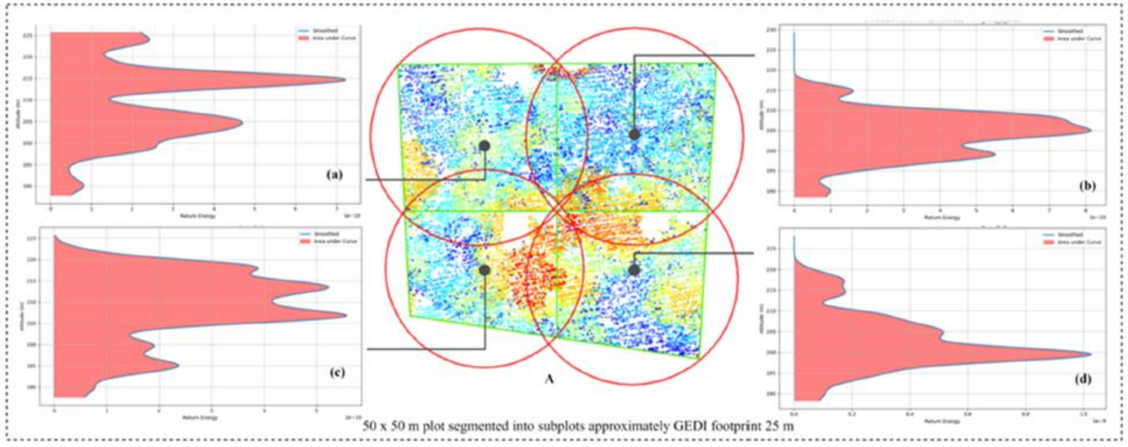
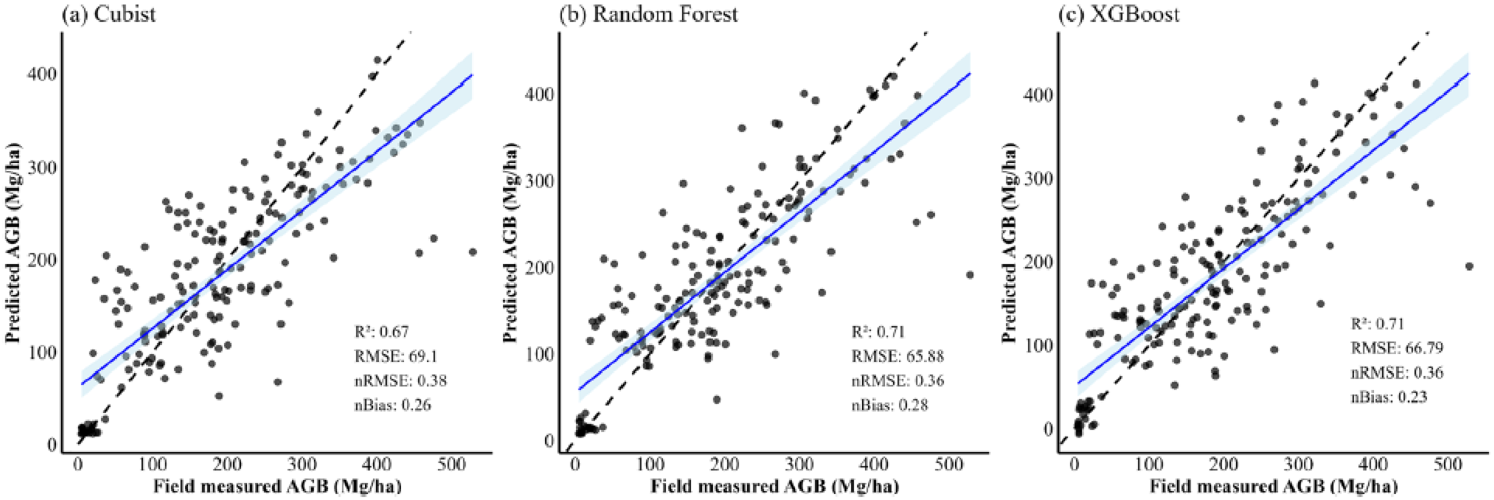


Fig. 6. A. ALS point clouds within a 50 × 50 m plot segmented into subplots of size ~25 m conforming to the GEDI footprints. (a – d) are corresponding simulated GEDI waveforms at the subplot level. Red circular footprints present the GEDI footprints and grey dots present the centroid of each circle.



Fareed et al. (2025)

Fig. 9. Predicted versus field measured AGB (Mg/ha) at plots (n = 174) using ML models, (a) Cubist, (b) RF, and (c) XGBoost. An extensive suite of mRH metrics was used for the model's evaluation and the descriptive statistics of RMSE, NRMSE, and nBais are provided in (Table 3).

GEDI-like full waveform simulation and extraction of RHs from ALS

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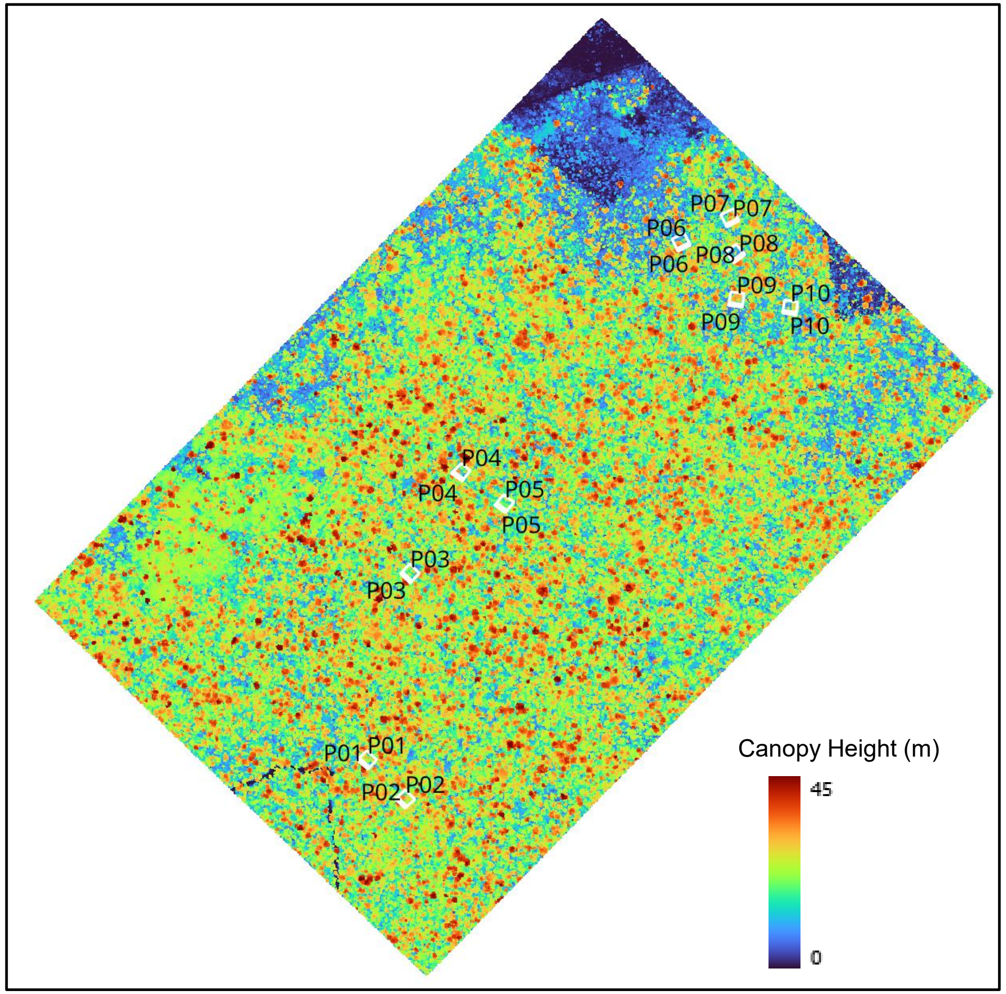
Exercise 1: GEDI Waveform Simulation

1. Simulate GEDI-like Waveforms from airborne lidar data for 10 field plots in Acre using GEDI Simulator **rGEDI**.
2. Extract Relative Height Metrics from simulated GEDI WVs.

- GEDI Simulator: **rGEDI** in R
  - Developed by Silva at el (2020)
  - Link: https://rgedi.r-forge.r-project.org/



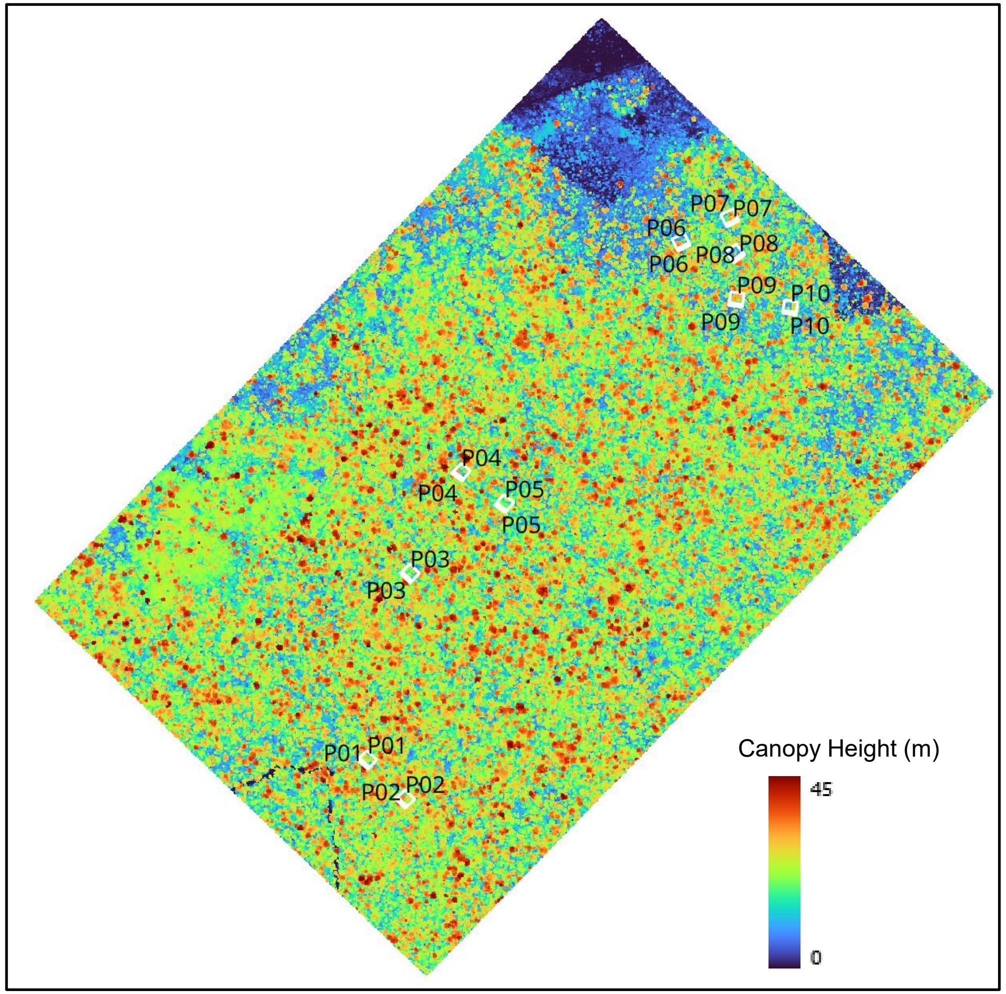
Bonal Forest Reserve in Acre

Data Source: https://www.earthdata.nasa.gov/data/catalog/ornl-cloud-lidar-forest-inventory-brazil-1644-1

GEDI-like full waveforms simulation and extraction of RHs from ALS

Exercise 1

- Area: Bonal Reserve in Acre, Brazil.
- Data: Forest inventory and ALS data from Sustainable Landscape Project (https://www.earthdata.nasa.gov/data/catalog/ornl-cloud-lidar-forest-inventory-brazil-1644-1).
- Forest inventories in 2013 and ALS flight in 2014.
- Field Plot Size: 50m x 50m
- Input data (airborne data: las):
  - **/Exercise_1/plot_las/**
- Output folder:
  - **/Exercise_1/output01/**



Bonal Forest Reserve in Acre

Data Source: https://www.earthdata.nasa.gov/data/catalog/ornl-cloud-lidar-forest-inventory-brazil-1644-1

GEDI full waveforms simulation from ALS

1. Open “*rGEDI_sim_withplots_exercise_1.r*” script in Rstudio.

2. Note that the following r-packages

   library(rGEDI)
   library(rGEDIsimulator)
   library(lidR)
   library(plot3D)

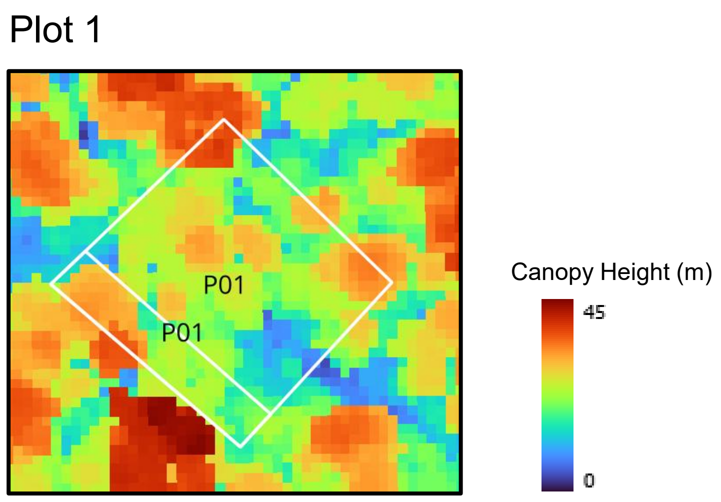
3. Define paths for “las_folder” and “output_folder” in the script.
   - Las_folder: *‘X:/xxx/**Exercise_1/plot_las**’*
   - Output folder: *‘X:/xxx/**Exercise_1/output01**’*

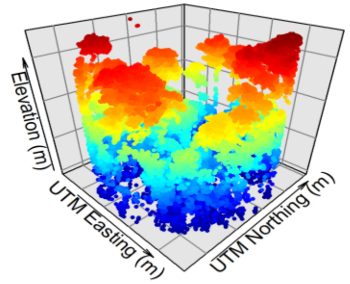
4. Clip ***Ctrl+Alt+R*** to run the script**.**

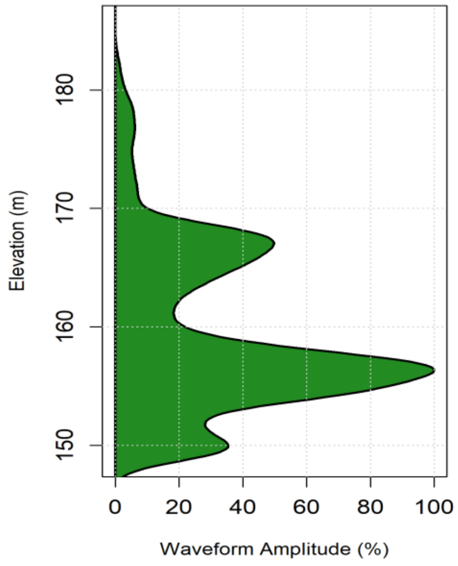
5. Simulation function: “**gediWFSimulator**”

   # Simulate GEDI waveform
   *wf <- **gediWFSimulator**(input = las_file, output = wf_file, coords = coords)*

Verify the *output WV files in H5 format* and the *figures of point-cloud and GEDI-like Waveform* of the study plots in “output01” folder.







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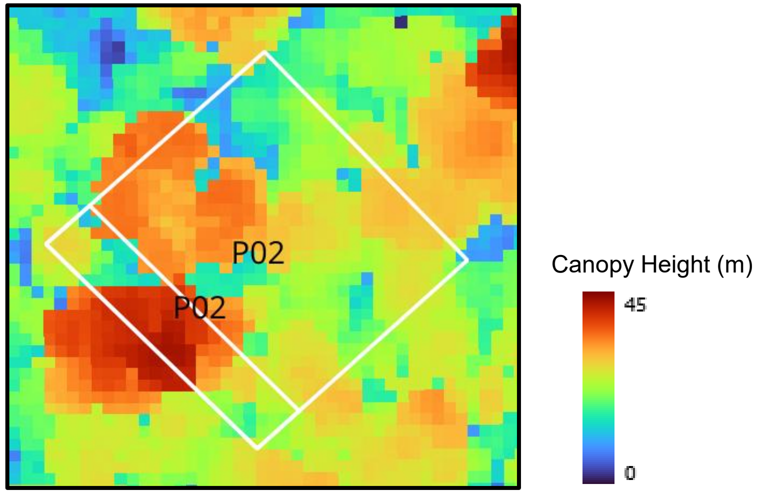
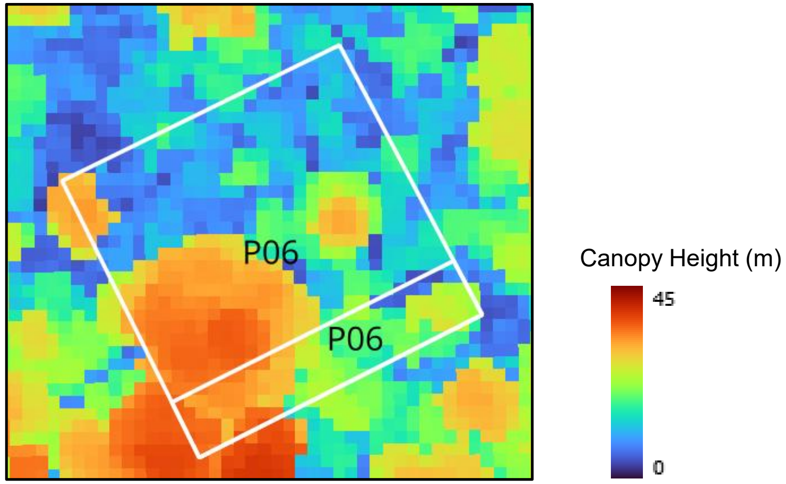
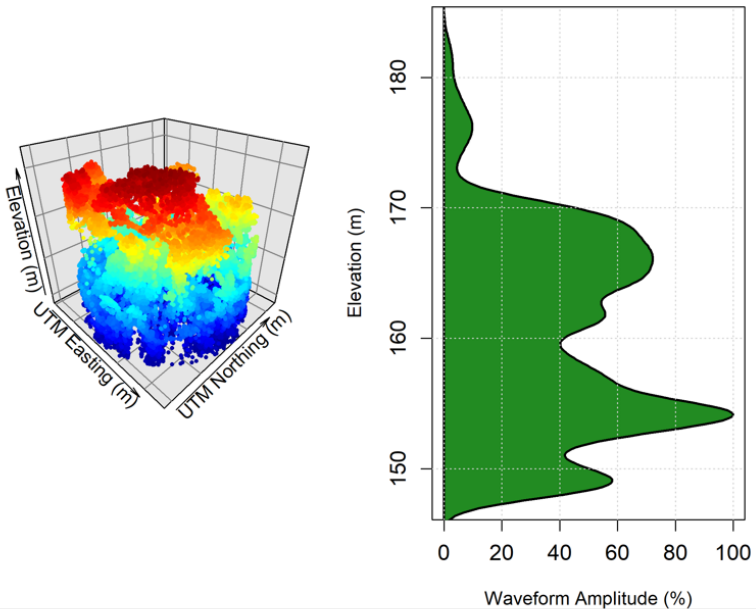
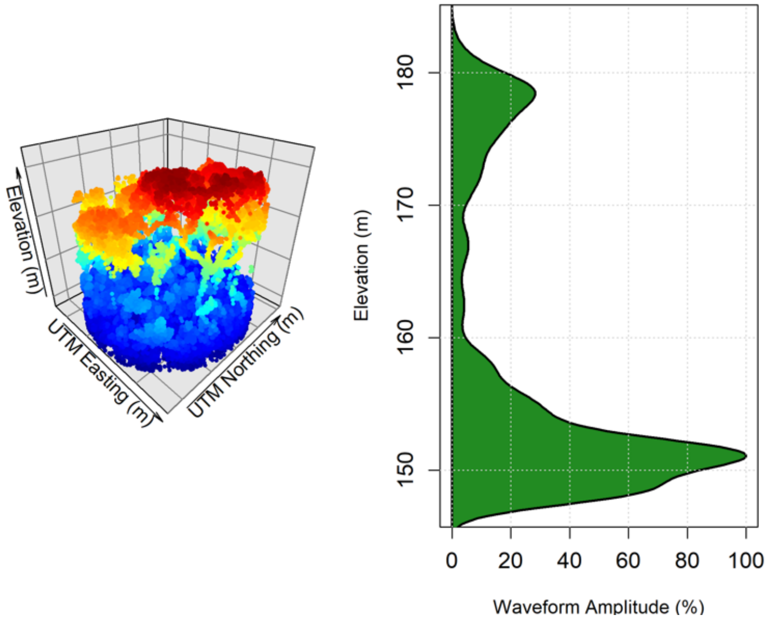








Extraction of RH metrics from simulated GEDI waveforms

Mean RHs from multiple WVs for field plot

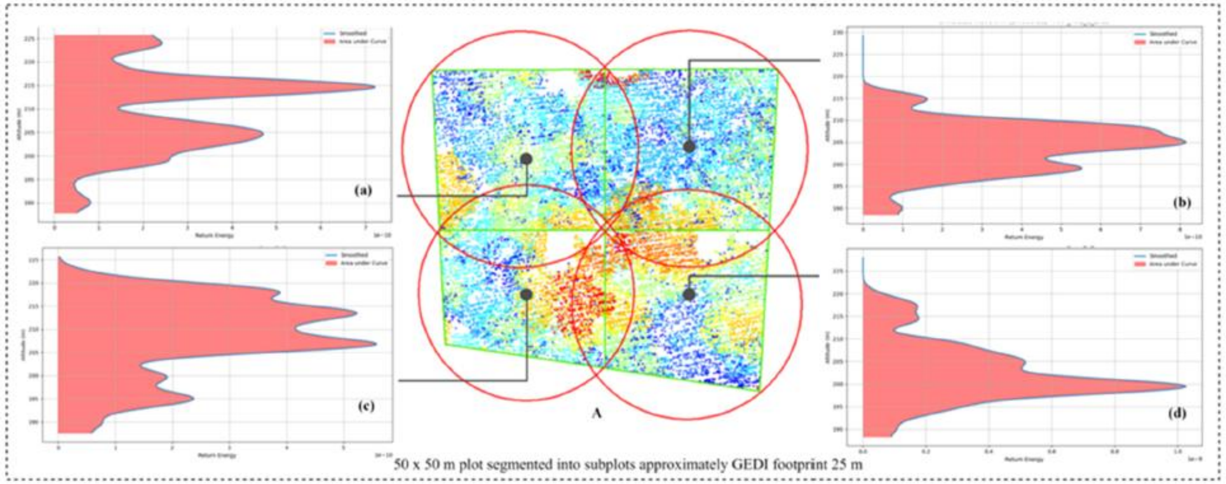
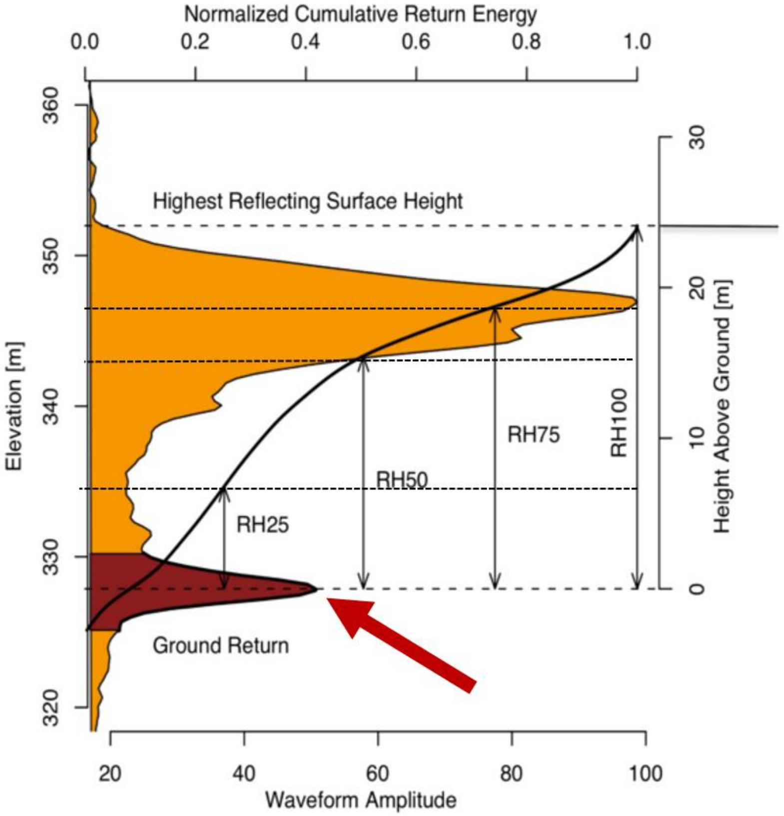


Fig. 6. A. ALS point clouds within a 50 × 50 m plot segmented into subplots of size ~25 m conforming to the GEDI footprints. (a – d) are corresponding simulated GEDI waveforms at the subplot level. Red circular footprints present the GEDI footprints and grey dots present the centroid of each circle.

Extraction of Relative Height Metrics from simulated GEDI full waveforms



Extraction of RHs from simulated GEDI waveforms

- Extract RHs from simulated GEDI WF
  - Run script: “RH_calculation_from_rGEDI_h5_files.r”

  - *gediWFMetrics function for RH calculation*

    *wf_metrics<-**gediWFMetrics**(input=wf_h5, outRoot=file.path(getwd(), “output01"))*

  - The output file shows several forest structure measurements including Relative Heights (RHs).
    - ex: “BON_A01_3_5_gediWF_simulation_RH_metrics.csv”

rhGauss 9	rhGauss 9	rhGauss 1	RH0	RH5	RH10	RH15	RH20	RH25	RH30	RH35	RH40	RH45	RH50	RH55	RH60	RH65	RH70	RH75	RH80	RH85	RH90	RH95	RH100	rhInfl 0	rhInfl 5	rhInfl 10	rhInfl 15	rhInfl 20
21.26	23.36	31.76	-7.35	-2.7	-1.5	-0.45	0.6	1.5	2.55	3.3	4.05	4.8	5.7	7.05	9	11.25	13.5	14.7	15.9	16.8	18	20.1	28.5	-4.37	0.28	1.48	2.53	3.58

Developing local AGB model

Once you have RH metrics from simulated GEDI WFs and associated field biomass for all your plots, you can now develop AGB prediction model by ML algorithms.

Field measurements and ALS (UAV) lidar data collection

⇓

GEDI full-waveform simulation from ALS

⇓

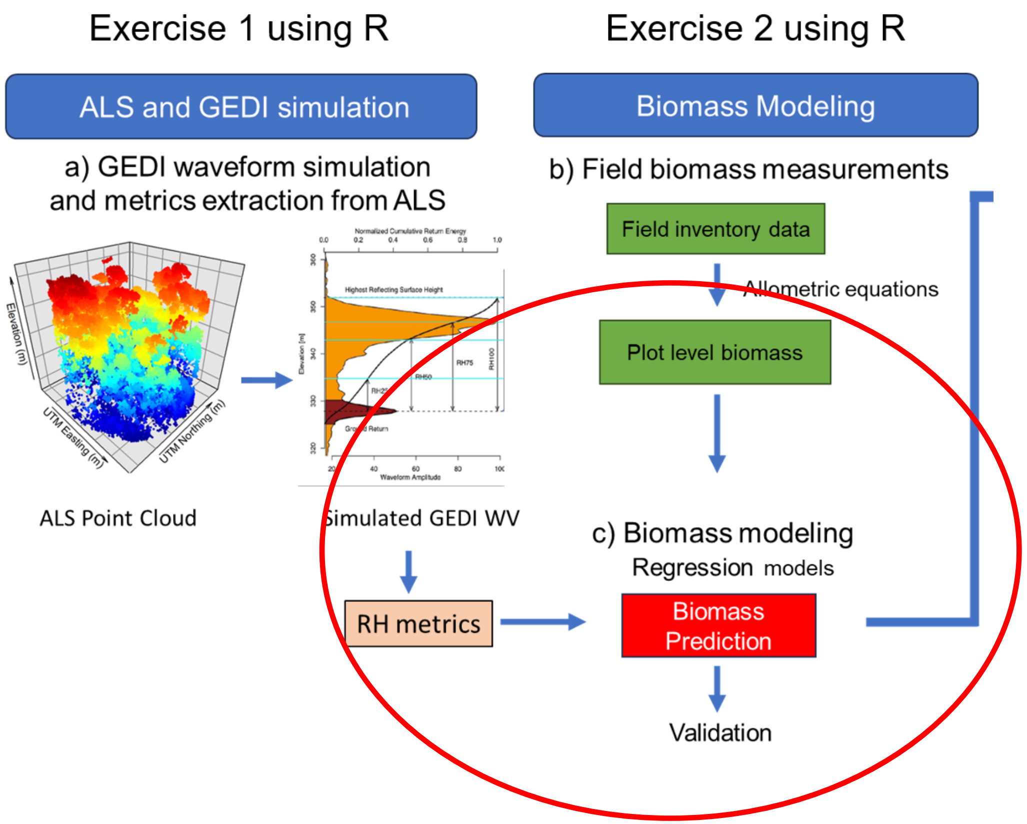
Extraction of Relative Height Metrics from simulated GEDI WVs

⇓

Development of AGB models using field AGB and RH metrics

⇓

Validation



Exercise 2: AGB Modeling

Exercise 2: Develop Random Forest based AGB Model in R

- Input file name: “***AGB_simulated_GEDI_RHs.csv***” in ***Exercise_2*** folder.
- In total, there are 173 plots from Sustainable Landscape Project.
- Response variable: “AGB”
- Predictor variables: “RH5….., RH100”
- Method: Random Forest + Cross Validation
- Reference: Fareed et al (2025), *Forest Ecology and Management*.

AGB	RH5	RH10	RH15	RH20	RH25	RH30	RH35	RH40	RH45	RH50	RH55	RH60	RH65	RH70	RH75	RH80	RH85	RH90	RH95	RH100
183.81	1.16	3.57	5.58	5.95	7.32	8.82	8.04	8.45	9.50	10.90	12.23	13.63	14.98	16.15	17.42	19.38	22.02	24.08	26.65	35.87
254.85	0.70	1.84	3.54	4.33	5.40	6.35	7.38	8.45	9.77	11.18	12.43	13.87	16.50	18.15	19.75	21.27	23.05	24.63	26.30	35.33
271.75	0.35	2.40	5.01	5.01	6.41	7.65	8.79	8.74	9.77	11.04	12.26	13.39	14.49	15.64	17.06	18.49	19.86	21.15	22.89	32.34
329.87	1.54	4.82	7.43	9.37	10.88	12.30	13.53	14.68	15.88	17.23	19.05	20.75	22.30	23.78	25.43	27.25	29.03	31.22	33.33	42.48
231.21	0.48	2.40	4.22	6.30	7.16	8.70	9.02	10.42	11.70	12.97	14.20	15.38	16.73	18.37	19.75	21.32	23.15	25.80	28.45	36.03
111.16	0.00	0.00	0.00	0.00	0.10	0.52	0.73	1.16	1.85	3.57	4.83	7.70	11.32	13.37	16.72	19.38	21.47	23.60	25.68	34.42
179.41	0.00	0.00	0.00	0.38	0.66	1.22	1.72	2.53	3.55	4.72	6.02	7.47	8.85	10.17	11.67	13.78	16.18	18.10	20.42	29.18
189.32	0.00	1.50	1.80	2.89	2.38	3.45	4.76	5.53	7.00	8.52	10.02	11.43	13.08	15.38	17.23	18.72	20.07	21.60	23.33	31.37
113.47	0.00	0.55	1.20	1.88	2.72	2.66	3.39	4.67	5.92	6.15	7.22	8.82	10.28	11.73	14.02	15.82	17.92	19.75	21.83	32.03
180.26	0.00	0.55	1.00	1.65	2.64	3.75	4.39	5.55	5.85	6.95	8.40	9.83	11.13	12.38	13.67	15.42	17.90	20.42	23.33	33.70
273.36	6.36	10.99	11.83	13.50	14.78	15.88	15.12	16.42	17.42	18.20	18.95	19.70	20.53	21.37	22.20	23.10	23.97	24.88	25.98	34.67
256.15	2.40	4.12	6.53	8.25	9.60	10.83	12.18	13.62	14.78	16.08	17.05	17.83	18.65	19.45	20.27	21.08	22.07	23.15	24.80	33.07
190.59	0.40	1.83	3.35	5.65	6.90	9.17	9.82	9.80	10.72	11.65	12.58	13.53	14.62	15.63	16.67	17.82	19.17	20.48	21.90	30.60
332.29	3.92	8.10	11.33	14.12	16.32	17.67	18.78	19.77	20.57	21.25	21.90	22.47	23.07	23.77	24.63	25.45	26.37	27.48	29.20	39.37
229.95	7.52	10.90	13.42	15.17	16.46	17.53	18.51	19.47	20.35	21.26	22.10	22.87	23.71	24.55	25.86	27.38	28.57	29.85	31.72	36.80
285.58	7.10	9.96	14.08	14.43	15.88	16.95	17.82	18.60	19.33	20.07	20.82	21.48	22.27	22.92	23.53	24.12	24.80	25.60	26.78	35.62
425.99	6.67	12.68	16.22	18.48	20.22	21.52	22.63	23.57	24.37	25.13	25.88	26.53	27.18	27.80	28.48	29.37	30.37	31.40	32.77	42.78
350.71	3.09	7.22	11.02	13.70	16.22	18.18	20.13	21.58	22.62	23.57	24.40	25.18	25.90	26.65	27.32	28.07	28.93	30.02	31.35	39.00
367.80	6.58	10.47	13.08	14.83	16.40	17.87	19.32	20.55	21.53	22.47	23.32	24.07	24.80	25.62	26.42	27.40	28.55	30.07	31.72	39.97
223.49	4.76	6.86	7.93	9.57	11.07	12.27	13.42	14.33	15.23	16.17	17.23	18.50	19.87	21.02	22.08	22.97	23.88	24.93	26.23	34.27

AGB in Mg/ha
RHs in m

AGB Modeling/Cross-validation

Open a text file “RF_model.txt” in Exercise_2 folder.

```
library(data.table)
library(randomForest)
library(caret)

rf_input = read.csv(“X:/XXXX/Exercise_2/AGB_simulated_GEDI_RHs.csv")

# Set up K-fold cross-validation
set.seed(123)                      # for reproducibility
ctrl <- trainControl(
method = "repeatedcv",             # Repeated K-fold cross-validation
number = 10,                       # Number of folds
repeats = 3,                       # Number of repetitions
savePredictions = "final")         # Save predictions for analysis

# Perform cross-validation
rf_model_cv <- train(AGB ~ .,      # Formula: outcome as a function of all other variables
data = rf_input,
method = 'rf’,
ntree = 100,
trControl = ctrl,
tuneLength = 5)
```

RF model/CV Results

> print(rf_model_cv)

Random Forest

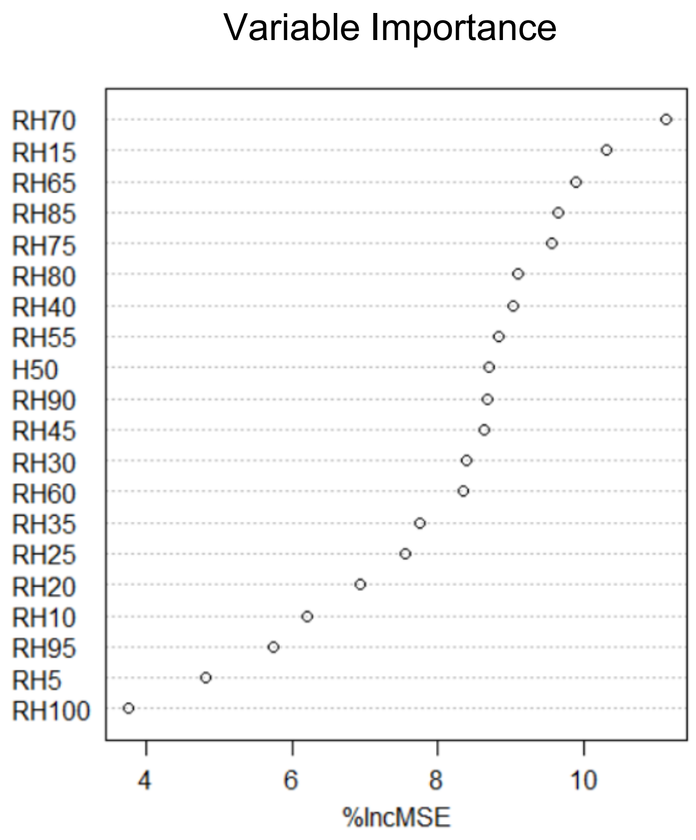
173 samples
 20 predictor

No pre-processing
Resampling: Cross-Validated (10 fold, repeated 3 times)
Summary of sample sizes: 153, 155, 156, 156, 155, 156, ...
Resampling results across tuning parameters:

mtry	RMSE	Rsquared	MAE
2	64.32955	0.7128399	46.77447
6	64.60245	0.7109242	47.04979
11	65.20188	0.7050827	47.52374
15	65.18110	0.7059738	47.64155
20	65.35264	0.7025980	47.67170

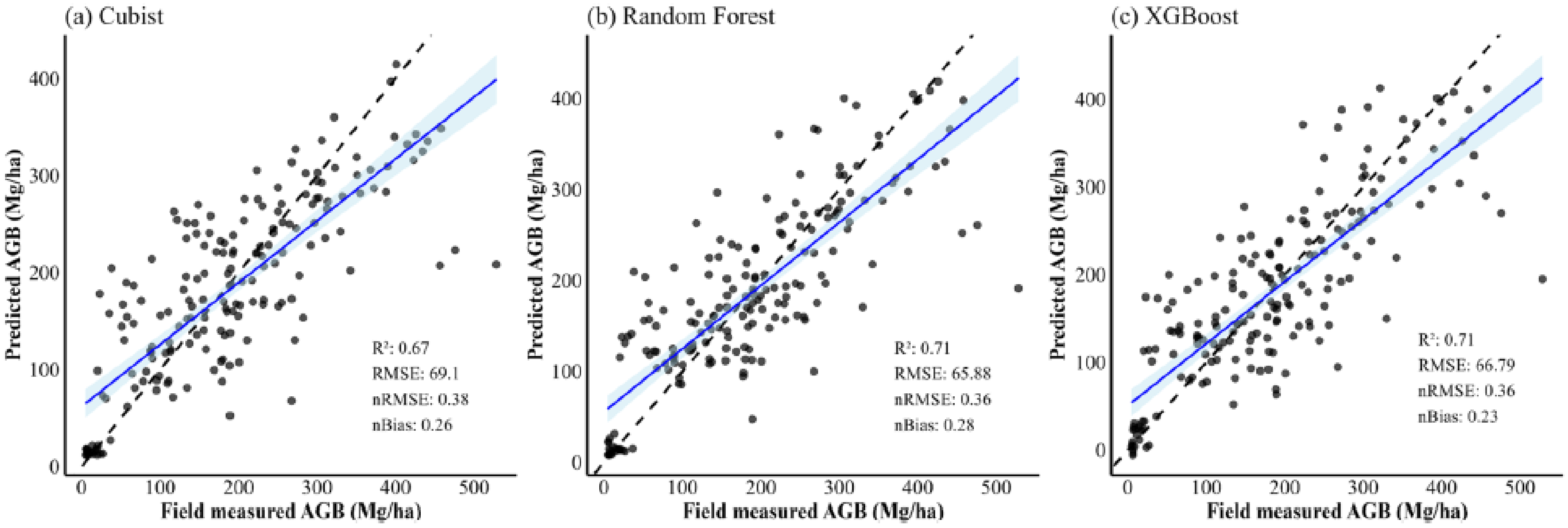
RMSE was used to select the optimal model using the smallest value.
The final value used for the model was mtry = 2.

** "mtry" stands for the number of variables randomly sampled as candidates at each split when growing a tree.*



AGB Modeling

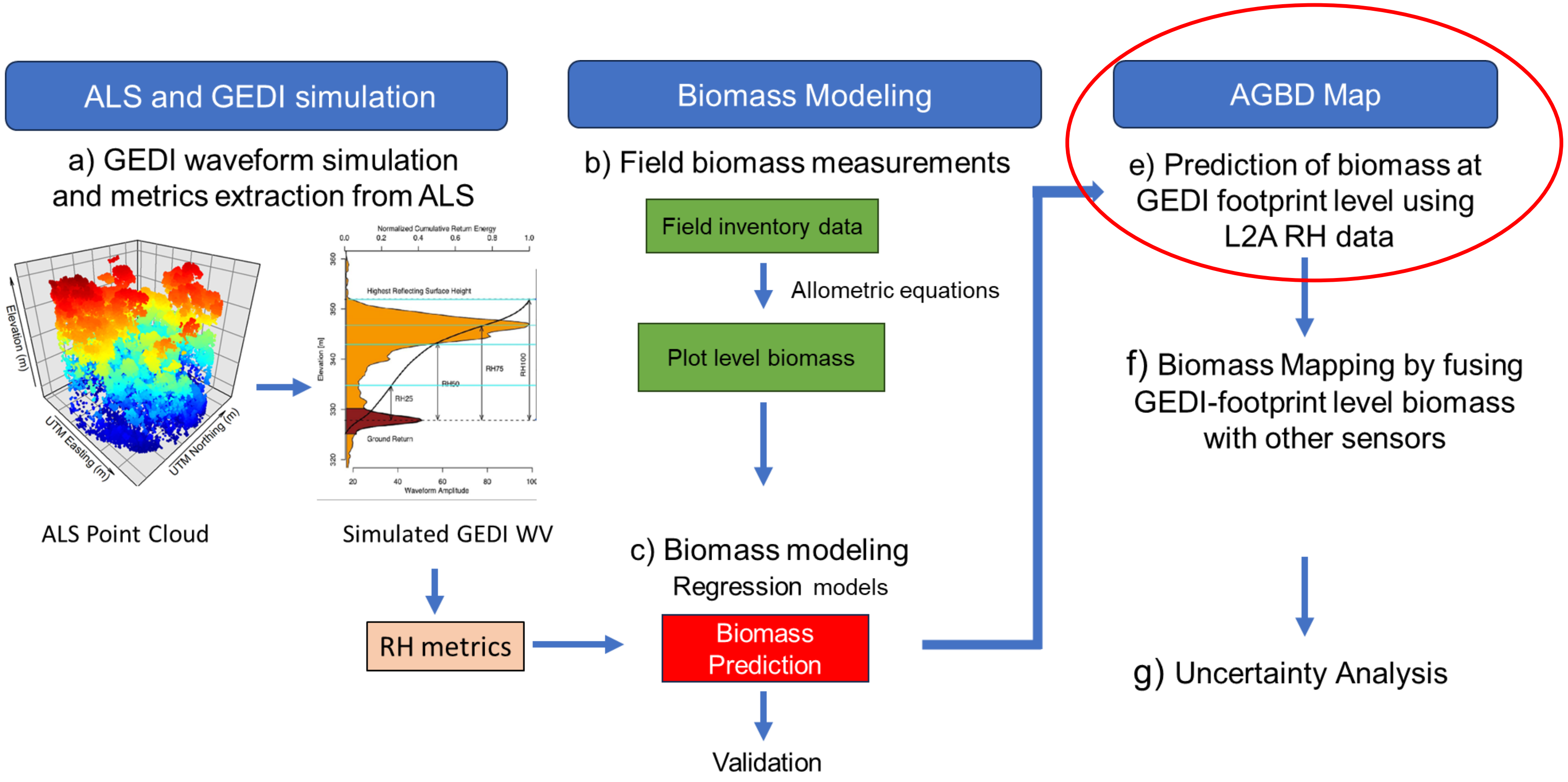
ML models for AGB prediction



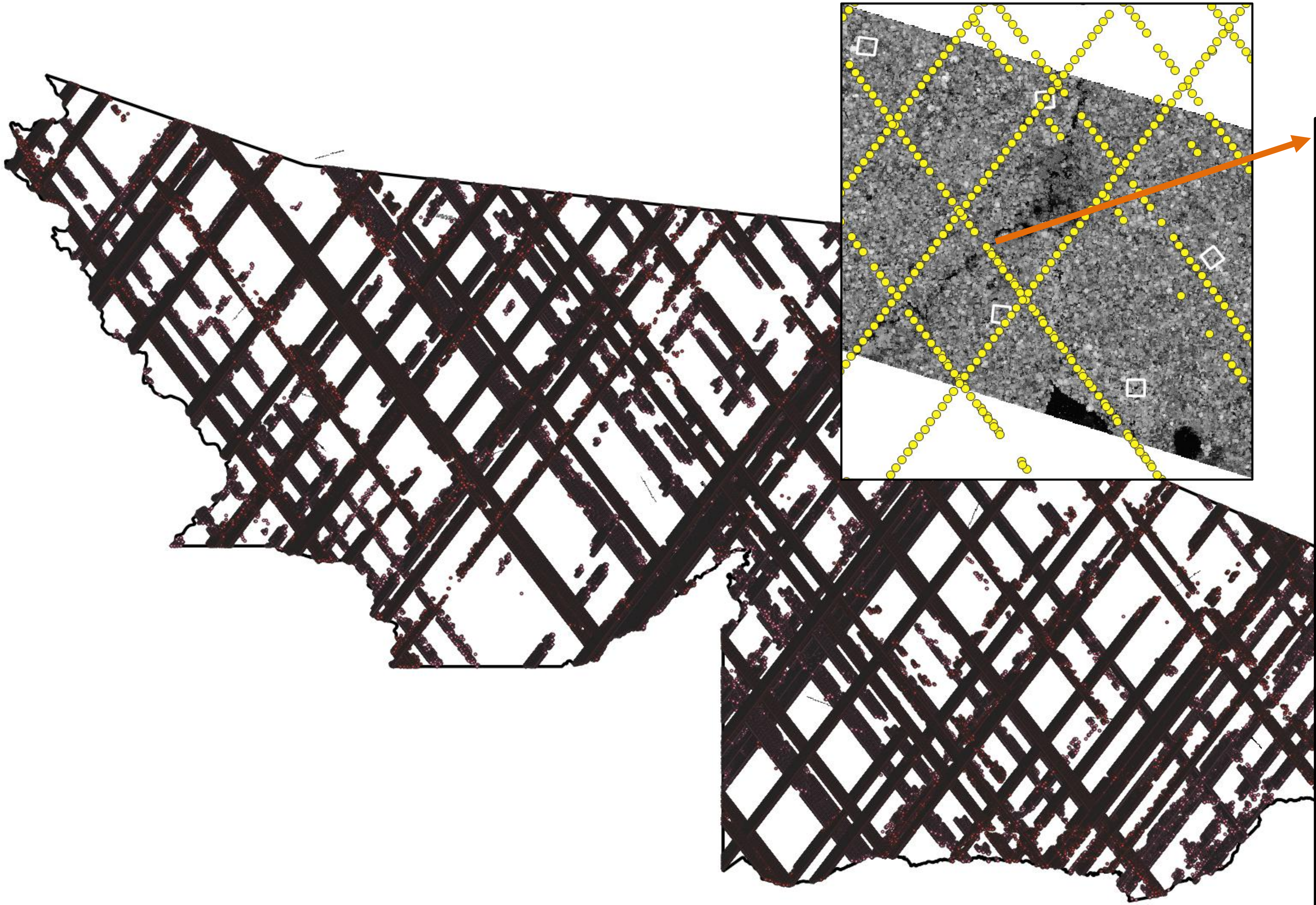
Fareed et al. (2025), Forest Ecology and Management

Apply LOCAL AGB model to GEDI data

- Once we have local AGB model, we can apply this to GEDI RH Metrics data to predict footprint level AGB data over a large area.
- GEDI footprint based AGB data can be used for wall-to-wall AGB model through fusing with satellite data like Landsat.



GEDI L2A Elevation and RH Metrics



Points from table

field_1	19683
(Derived)	
(Actions)	
field_1	19683
File	20242100
Beam	BEAM1011
RH05	-3.17
RH10	0.55
RH15	4.21
RH20	7.27
RH25	9.55
RH30	11.12
RH35	12.65
RH40	14.25
RH45	15.71
RH50	17.27
RH55	18.54
RH60	19.51
RH65	20.52
RH70	21.6
RH75	22.5
RH80	23.28
RH85	24.1
RH95	25.97
RH98	26.79
RH100	28.66
latitude	-7.97726865436397
longitude	-72.02847381644341
DEM	234.83537
g_elev	221.60538
sensitivity	0.98850167
s_elevation	-75.6661
deg_flag	0
n_mode	8
algorithm	2
shot_n	318591100400375236

GEDI footprint selection

- Some criteria for GEDI footprint selection:
  - Quality Flag: This flag indicates usable data by summarizing individual quality assessment parameters (e.g., waveform shot energy, sensitivity, amplitude, etc).
  - Bean Sensitivity: >=0.90 or higher sensitivity for dense forest.
  - Solar Elevation : > 0 at daytime and < 0 at nighttime.
  - Beams: 4 cover beams (low penetration) and 4 power beams (high penetration).
  - Elevation Difference: GEDI ground elevation vs. TanDEM-X
    - Threshold values vary… For tropical forests, 100-150m (?)
  - Anomalous Value Removal: Remove footprints with extreme or unrealistic values for metrics like canopy height or aboveground biomass density (AGBD).

Apply RF AGB model to GEDI L2A RH Metric data

- Load “filtered” 2019 GEDI L2A data for Acre.

> gedi_ac <- read.csv(“X:/XXXX/Exercise_2/GEDI_2A_ACRE_2019_filtered.csv")

Apply RF model to GEDI L2A data with the same RHs.

> gedi_predict <- predict(**rf_model_cv**, newdata = **gedi_ac**)

Concatenate gedi_predict with footprint coordinates (“latitude” and “longitude”) from ‘gedi_ac’

> gedi_ac$AGB <- gedi_predict

> gedi_agb <- gedi_ac[, c(24:25, 27)]

Select randomly 5,000 AGB footprints from this data for GEE AGB prediction model as GEE can not handle a large data (5,000 elements at max).

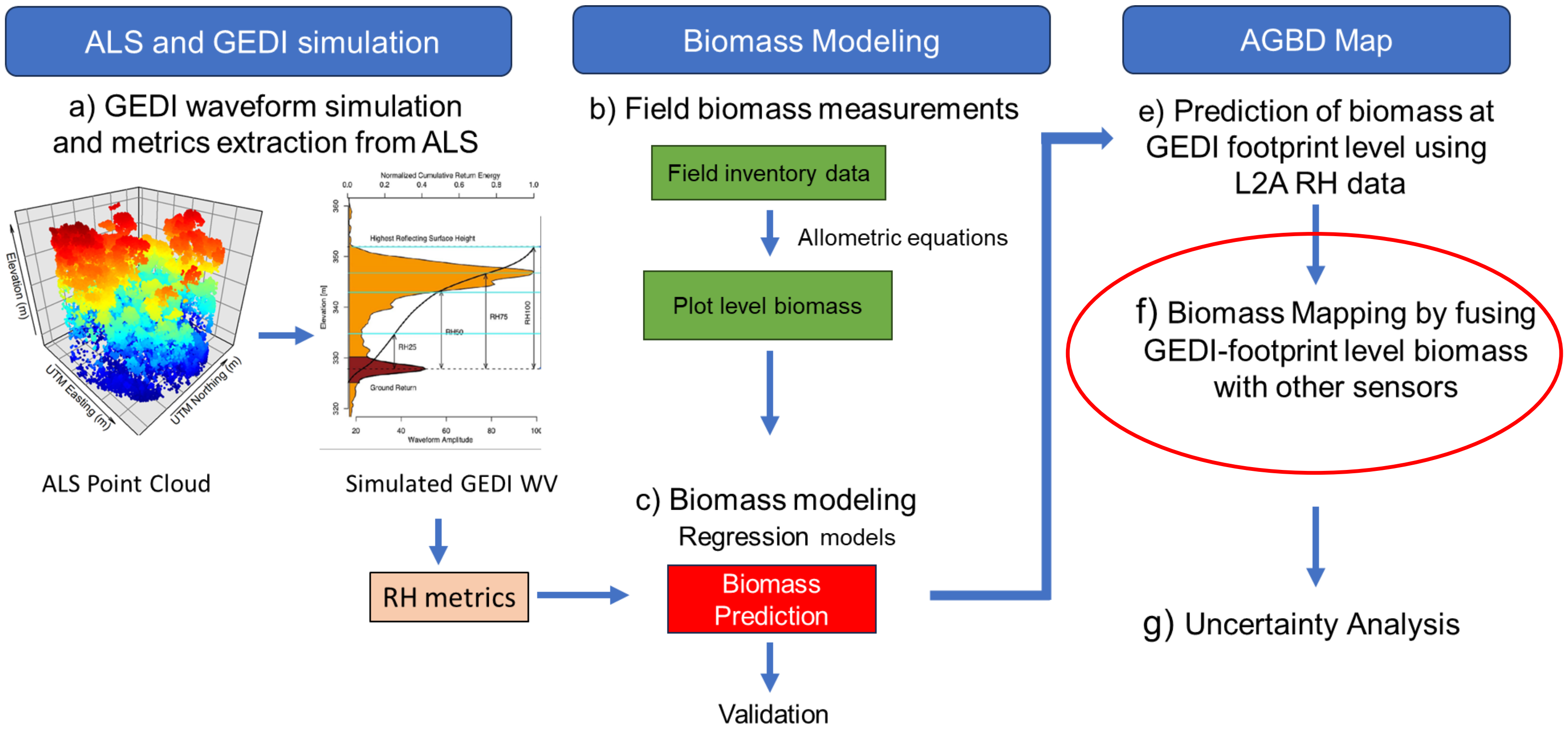
> sampled_agb_index <- sample(1:nrow(gedi_agd), size = 5,000, replace = FALSE)

> sampled_agb <-gedi_agb[sampled_agb_index,]

> write.csv(sampled_agb, ‘X:/XXXX/Exercise_3/gee_agb_ac.csv')

AGB Mapping

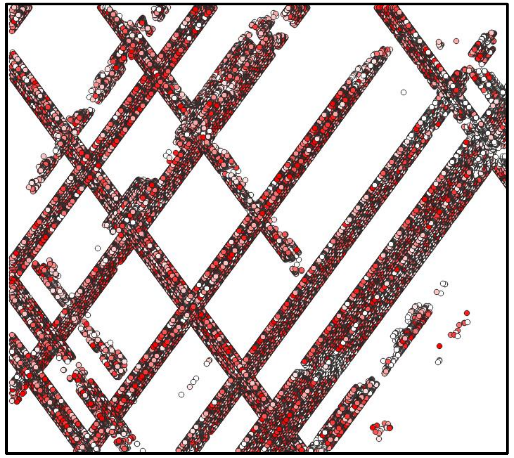
- Once we have local AGB model, we can apply this to GEDI RH Metrics data to predict footprint level AGB data over a large area.
- GEDI footprint based AGB data can be used for wall-to-wall AGB model through fusing with satellite data like Landsat.



Annual AGB Mapping by fusing GEDI AGB with other Remote Sensing data

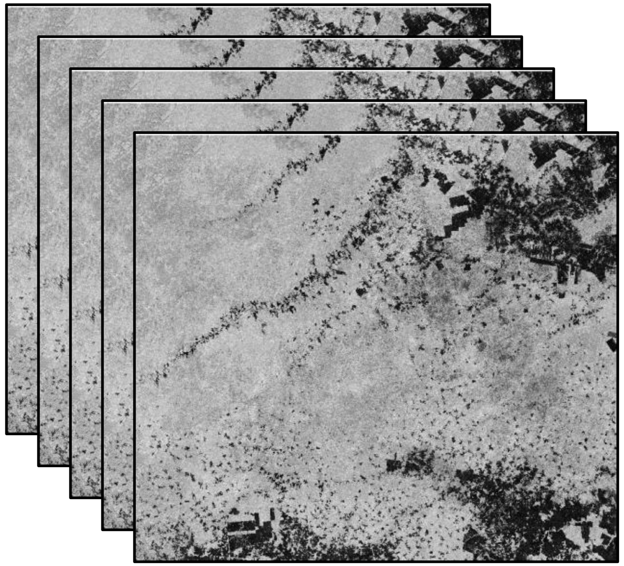
SERVIR

Response Variable
GEDI-footprint AGBD

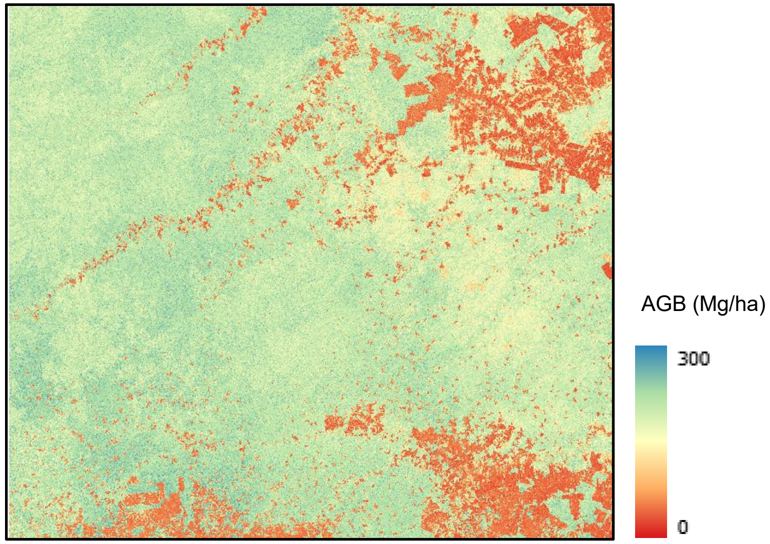


ML Model (RF)
for upscaling
AGB

Predictor Variables
Landsat/Sentinel/DEM/Radar
~~- Temporal Composite Metrics~~
- CCDC Coefficients and Synthetic image



AGBD Map



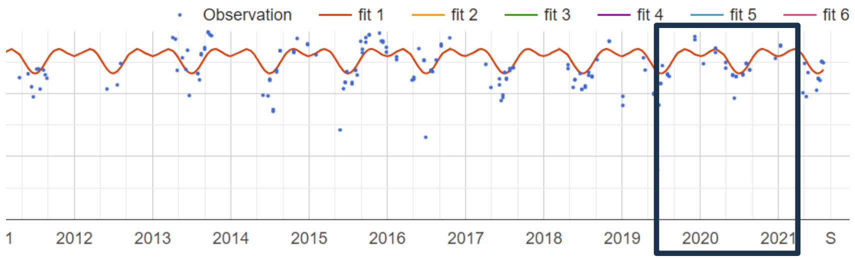
Exercise 3: AGB mapping using GEE

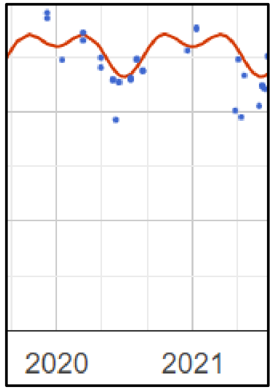
- AGB mapping using **Google Earth Engine**
- ML model: Random Forest
- AGB data: GEDI AGB predicted by the model we developed in Exercise 2
- Predictor Variables from Landsat (and Sentinel)
    - Temporal Composite/Statistical Metric (e.g., mean, median, stDev, percentailes, etc)
    - Continuous Change Detection Classification (CCDC) – Coefficients and Synthetic Image
    - Other predictor variables: DEM and Slope from Copernicus Glo30

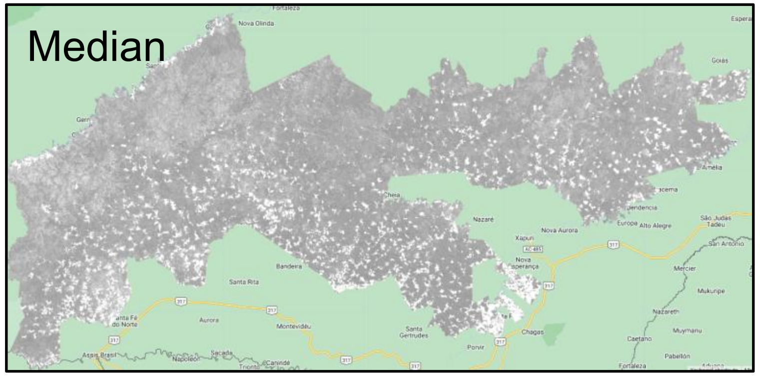
Landsat temporal composite Metrics as predictor variables

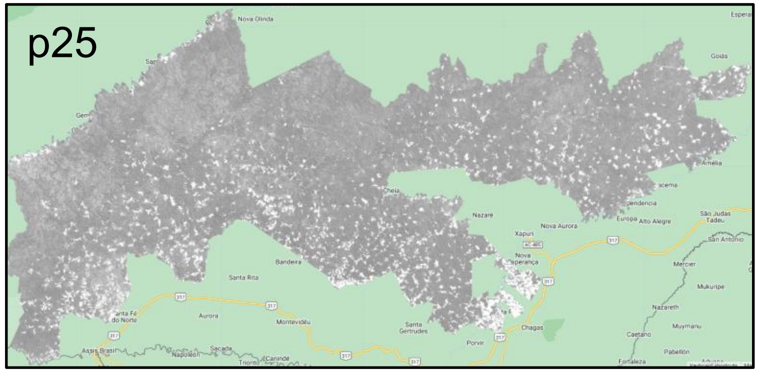
Landsat (or Sentinel) derived temporal composites (annual, quarter or seasonal, etc)

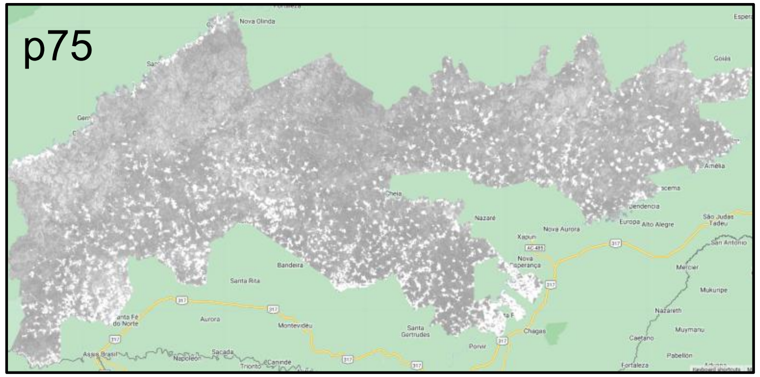
Metrics – Median, mean, stdev, max, min, p25, p75…











Issues with temporal composite/metric data

SERVIR

- Varying viewing geometry across year
- Varying climate conditions from year to year (ex: drought/non-drought year) affect vegetation responses.
- Varying image availability from year to year
- Spectral variability among image scenes
- Topography effect (hilly vs. flat surface)

•Continuous Change Detection Classification – CCDC

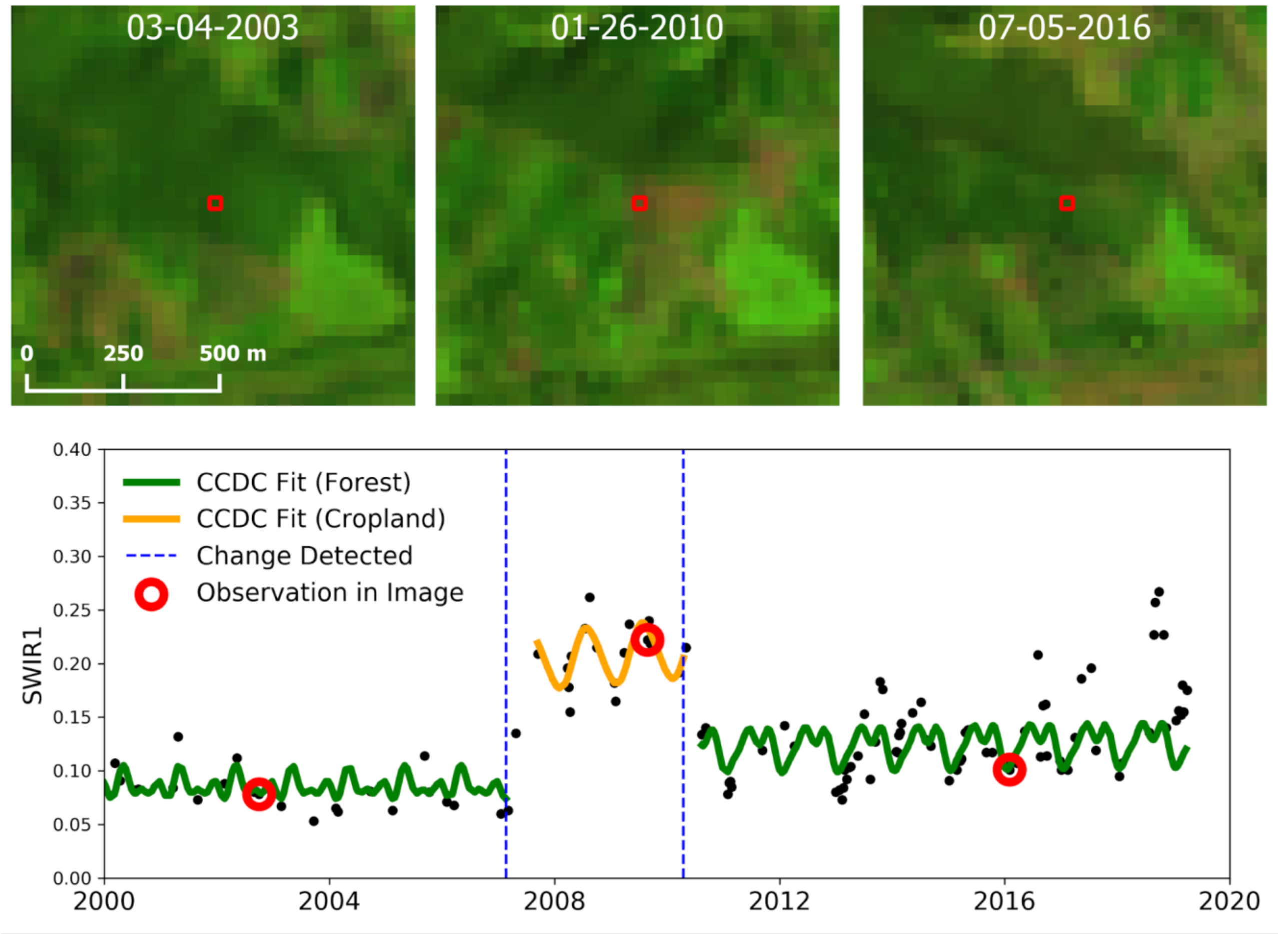
- CCDC Coefficients
- CCDC Synthetic Image

$$\hat{\rho}(i,x)_{OLS} = a_{0,i} + a_{1,i}\cos\left(\frac{2\pi}{T}x\right) + b_{1,i}\sin\left(\frac{2\pi}{T}x\right) + c_{1,i}x \qquad (3)$$

$$\{\tau^*_{k-1} < x \le \tau^*_k\}$$

where,

x	Julian date
i	the ith Landsat Band
T	number of days per year ($T = 365$)
$a_{0,i}$	coefficient for overall value for the ith Landsat Band
$a_{1,i}, b_{1,i}$	coefficients for intra-annual change for the ith Landsat Band
$c_{1,i}$	coefficient for inter-annual change for the ith Landsat Band
τ^*_k	the kth break points.
$\hat{\rho}(i,x)_{OLS}$	predicted value for the ith Landsat Band at Julian date x.



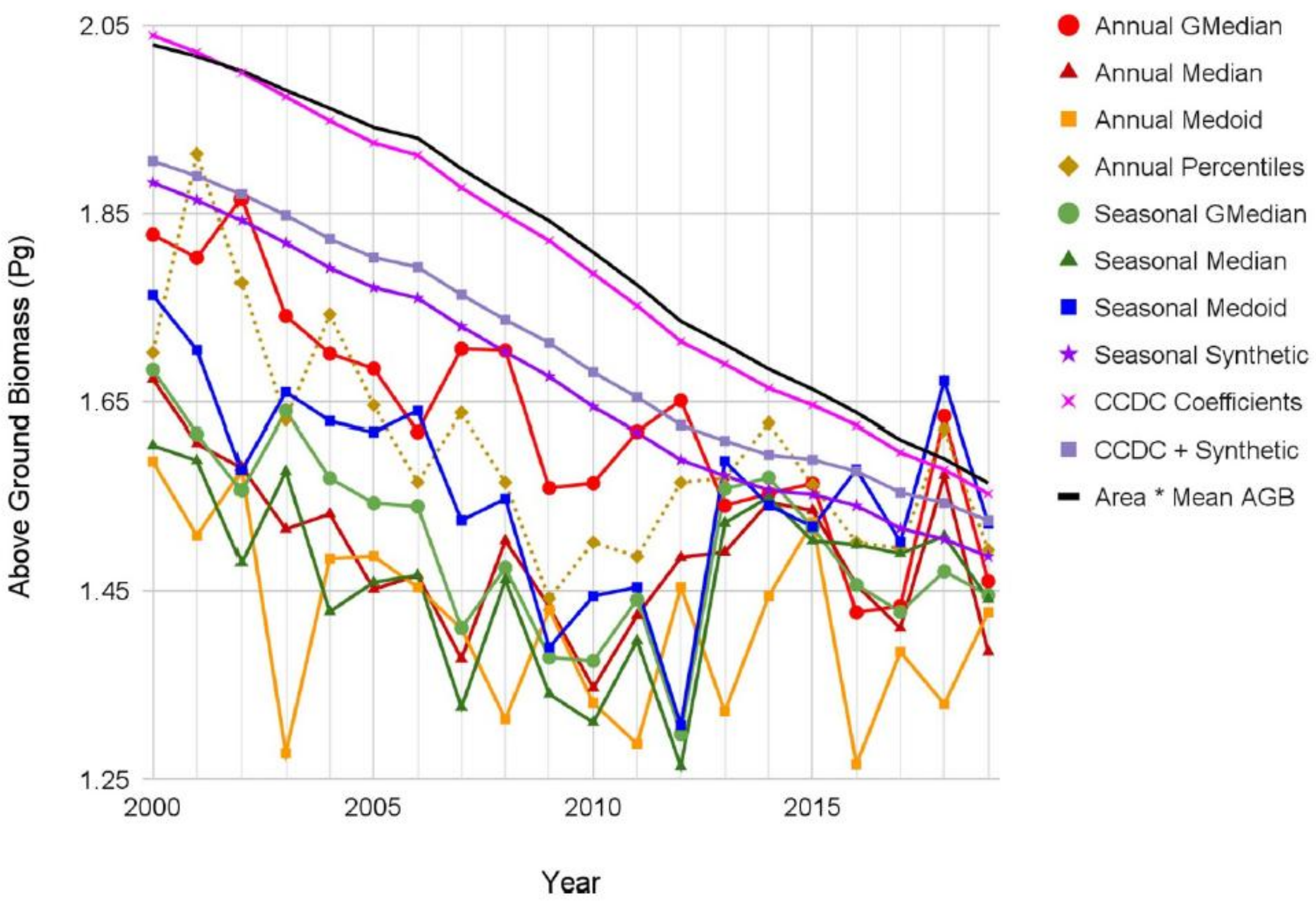


Figure 2. Above-ground biomass predictions for the forested regions of Paraguay, using ten different feature vectors. The black line shows the area of forest scaled by the mean biomass value from the reference data.

Gorelik et al (2023) in *ERL*

GEE

- AGB mapping using **Google Earth Engine**
- Scrips:
  - **GEDI_AGBD_prediction_CCDC**: for Subset image

    https://code.earthengine.google.com/dafc6a5d10ba20029d3af301e0f351d6

GEE

- AGB mapping using **Google Earth Engine**
- Scrips: (Links):
  - GEDI_AGBD_prediction_CCDC:
- Script Structure
  - Setting up Parameters
  - 1. Preparing Input data
    - 1.1. GEDI footprint AGB data
    - 1.2. CCDC
    - 1.3. DEM (Glo30)
  - 2. AGB Modeling
    - 2.1. Preparing sampling data
    - 2.2. RF modeling and Cross validation
    - 2.3. AGB mapping
  - 3. Exporting output data

Thank you!